

### Description about the data:

The data consists of 522 observations of Sale price of the home (Independent variable). The data explores the factors affecting the price of the house like square footage of home and square footage of lot (explanatory variables).

### Analysis Report:

#### Model

The regression model is given as follows:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2$$

Where Y=Price of the home

$X_1$ = Square footage of the home

$X_2$ = Square footage of the lot

#### To check the overall significance

$H_0: \beta_1 = \beta_2 = 0$  ; i.e the model is not significant

$H_1$ : At least one of the  $\beta_i$  is not equal to zero ; i.e the model is significant

Level of significance= 5%= 0.05

Since the p-value ( $2.2e-16$ ) is less than 0.05 we reject  $H_0$ . This implies at least one of the regression coefficients is not zero and the model is significant.

```
> model1=lm(data$`Sales Price`~data$`home ft2`+data$`lot ft2`)  
> summary(model1)
```

```
Residual standard error: 78070 on 519 degrees of freedom  
Multiple R-squared:  0.6808,    Adjusted R-squared:  0.6796  
F-statistic: 553.5 on 2 and 519 DF,  p-value: < 2.2e-16
```

#### To check the individual significance

$H_0: \beta_i = 0$ ;  $i=1,2,3$ . i.e the parameters are significant

$H_1: \beta_i \neq 0$ ; i.e the parameters are not significant

Level of significance= 5%= 0.05

```
> model1=lm(data$`Sales Price`~data$`home ft2`+data$`lot ft2`)  
> summary(model1)
```

```

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -1.027e+05  1.265e+04  -8.121 3.39e-15
data$`home ft2`  1.560e+02  4.871e+00  32.019 < 2e-16
data$`lot ft2`   1.151e+00  2.964e-01   3.882 0.000117

```

Since the p-value is less than 0.05 for all the parameters we can confirm that all of the coefficients are significant.

### Testing of the assumptions

#### 1) Autocorrelation

**To test the presence of autocorrelation we used Durbin-Watson Test**

$H_0$ :  $\rho=0$ ; i.e there is no autocorrelation

$H_1$ :  $\rho \neq 0$ ; i.e there is autocorrelation

Level of significance= 5%= 0.05

```

> durbinwatsonTest(model1)
lag Autocorrelation D-W Statistic p-value
1      0.3316004      1.336392      0
Alternative hypothesis: rho != 0

```

The p-value is zero which implies that autocorrelation is not present.

#### 2) Multicollinearity

**To test the presence of multicollinearity we used Variation Inflation Factors**

```

> vif_values=vif(model1)
> vif_values
data$`home ft2` data$`lot ft2`
      1.025445      1.025445

```

Since both the vif values are less than 10 we can say that multicollinearity is not present.

#### 3) Homoscedasticity

**To check the presence of heteroscedasticity we used Breusch-Pagan Test**

$H_0$ : Variance of the error terms is constant i.e the data is homoscedastic

$H_1$ : Variance of the error terms is significantly different i.e the data is heteroscedastic.

Level of significance= 5%= 0.05

```
> bptest(model1)
.
studentized Breusch-Pagan test

data: model1
BP = 85.21, df = 2, p-value < 2.2e-16
```

Since the p-value is less than 0.05 we reject our null hypothesis which means the given data is heteroscedastic.

As the data is not homoscedastic, we cannot use the OLS estimators. Hence, we performed the WLS (Weighted Least Square) method.

To find the weights we first had to calculate the variance of the error term. The code for the same is as follows:

```
est1=-1.027e+05+1.560e+02*data$`home ft2`+1.151e+00*data$`lot ft2`
variance=((data$`Sales Price`-est1)^2)/519
weights1=1/variance
```

After performing the WLS the model was as follows:

```
> model2=lm(data$`Sales Price`~data$`home ft2`+data$`lot ft2`,weights=weights1)
> summary(model2)

Call:
lm(formula = data$`Sales Price` ~ data$`home ft2` + data$`lot ft2`,
    weights = weights1)

Weighted Residuals:
    Min       1Q   Median       3Q      Max
-23.78 -22.77 -22.25  22.79  26.99

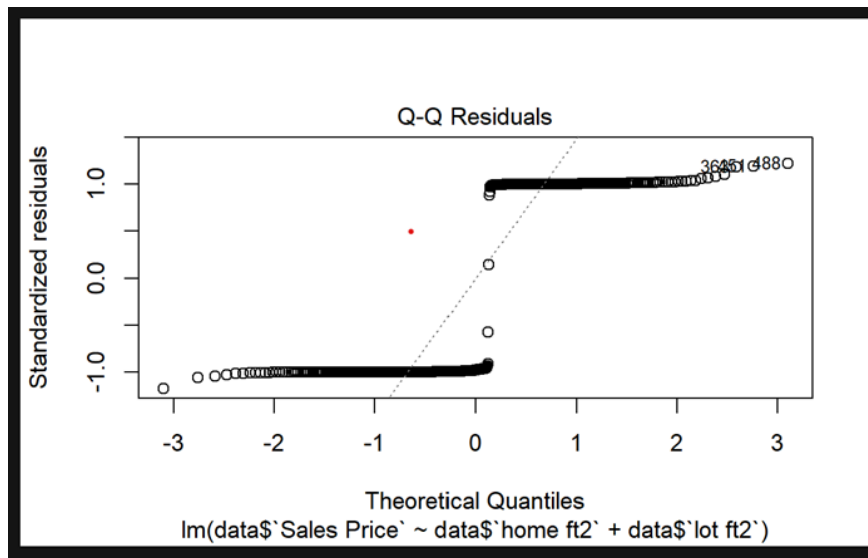
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) -1.032e+05  4.764e+02 -216.59  <2e-16 ***
data$`home ft2`  1.561e+02  1.441e-01 1083.56  <2e-16 ***
data$`lot ft2`   1.159e+00  2.567e-02   45.13  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 22.8 on 519 degrees of freedom
Multiple R-squared:  0.9996,    Adjusted R-squared:  0.9996
F-statistic: 7.121e+05 on 2 and 519 DF,  p-value: < 2.2e-16
```

As we can see from the t-test and F-test all the parameters are significant and the model is linear. The Adjusted  $R^2$  is 0.9966 which means 99.66% of the variation in the sales price of the home can be explained by the explanatory variables.

### Checking the assumptions:

#### 1) Checking for normality



The above graph shows that the residuals are not normally distributed.

**Hence to deal with this we used Log transformation of the dependent variable** (sale price of home).

After doing the log transformation we got the following model:

```

> model3=lm(log(data$`Sales Price`)~data$`home ft2`+data$`lot ft2`)
> summary(model3)

Call:
lm(formula = log(data$`Sales Price`) ~ data$`home ft2` + data$`lot ft2`)

Residuals:
    Min       1Q   Median       3Q      Max
-0.72796 -0.13706 -0.00891  0.11748  0.81345

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)   1.121e+01  3.726e-02  300.704 < 2e-16 ***
data$`home ft2` 4.989e-04  1.435e-05   34.774 < 2e-16 ***
data$`lot ft2`  4.167e-06  8.731e-07    4.772 2.37e-06 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.23 on 519 degrees of freedom
Multiple R-squared:  0.7173,    Adjusted R-squared:  0.7162
F-statistic: 658.5 on 2 and 519 DF,  p-value: < 2.2e-16

```

### Checking the assumptions:

#### 1) Homoscedasticity

**To check the presence of heteroscedasticity we used Breusch-Pagan Test**

$H_0$ : Variance of the error terms is constant ; i.e the data is homoscedastic.

$H_1$ : Variance of the error terms is significantly different; i.e the data is heteroscedastic.

Level of significance= 5%= 0.05

```

> bptest(model3)

studentized Breusch-Pagan test

data:  model3
BP = 46.272, df = 2, p-value = 8.959e-11

```

Here as observed the p-value is less than 0.05 hence we reject the null hypothesis. This suggests that the data is heteroscedastic and hence we perform the WLS method.

To find the weights the we used the following method :

```
est2=1.670157+0.098533*data$`home ft2`+0.009437*data$`lot ft2`  
variance2=((log(data$`Sales Price`)-est2)^2)/519  
weights2=1/variance2
```

After finding the weights WLS was performed to remove heteroscedasticity from the data

The transformed model is as follows:

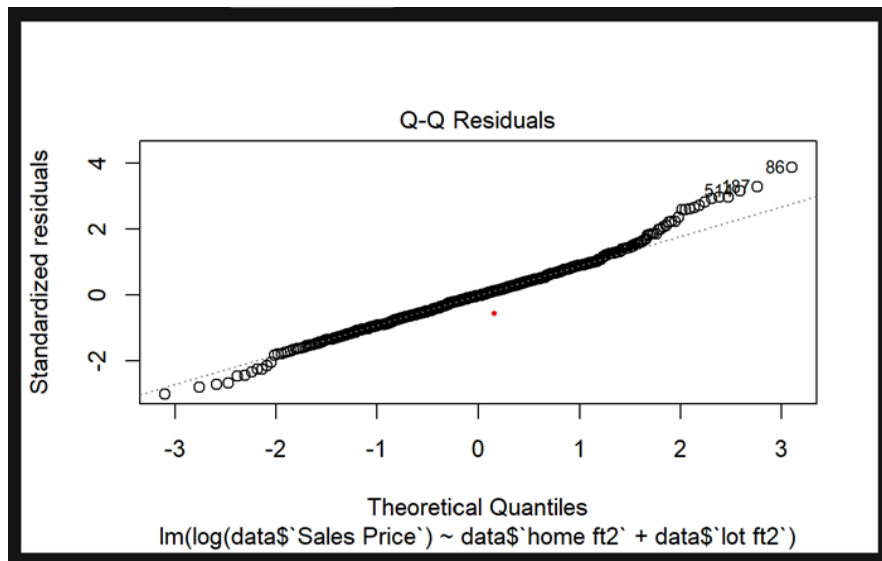
```
> model4=lm(log(data$`Sales Price`)~data$`home ft2`+data$`lot ft2`,weights=weights2)  
> summary(model4)  
  
Call:  
lm(formula = log(data$`Sales Price`) ~ data$`home ft2` + data$`lot ft2`,  
    weights = weights2)  
  
Weighted Residuals:  
      Min       1Q   Median       3Q      Max   
-0.035248 -0.007438 -0.000327  0.006741  0.045254  
  
Coefficients:  
              Estimate Std. Error t value Pr(>|t|)      
(Intercept)  1.112e+01  3.376e-02  329.501  < 2e-16 ***  
data$`home ft2` 5.241e-04  1.543e-05   33.964  < 2e-16 ***  
data$`lot ft2`  5.268e-06  1.101e-06    4.783  2.25e-06 ***  
---  
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1  
  
Residual standard error: 0.01176 on 519 degrees of freedom  
Multiple R-squared:  0.7245,    Adjusted R-squared:  0.7234  
F-statistic: 682.4 on 2 and 519 DF,  p-value: < 2.2e-16
```

The p-values of both the t-test and F-test suggest that all the parameters are significant and the model is linear.

The adjusted  $R^2$  value is 0.7234 which means 72.34% of the variation in the sales price of the home can be explained by the explanatory variables with  $\beta_1$  value as 5.421e-04 which implies that a unit increase in  $X_1$  (home ft.) variable will lead to an increase of 5.421e-04 in  $Y$  (sales price ) and with  $\beta_2$  value as 5.268e-06 which implies that a unit increase in  $X_2$  (lot ft.) variable will lead to an increase of 5.268e-06 in  $Y$  (sales price ) .

## Checking the Assumptions:

### 1) Checking for normality



As shown in the above graph there is a slight deviation from normality ; The adjusted  $R^2$  value decreased from 0.9966 to 0.7234 after log transformation which shows the decline in precision . To check the normality Shapiro-Wilk Test was performed and the results are as follows:

### Shapiro-Wilk Test

$H_0$ : Normality is present

$H_1$ : Normality is not present

Level of significance= 5%= 0.05

```
> shapiro.test(log(data$`Sales Price`)-est2)

      shapiro-wilk normality test

data:  log(data$`Sales Price`) - est2
W = 0.9124, p-value < 2.2e-16
```

The p-value is less than 0.05 which suggests that the error terms are not normally distributed which is the same as the conclusion that was drawn from the graph.

### 2) Homoscedasticity:

**To check the presence of heteroscedasticity we used Breusch-Pagan Test**

$H_0$ : Variance of the error terms is constant ; i.e, the data is homoscedastic

$H_1$ : Variance of the error terms is significantly different ; i.e, the data is heteroscedastic.

Level of significance= 5%= 0.05

```
> bptest(model4)

          studentized Breusch-Pagan test

data:  model4
BP = 0.2939, df = 2, p-value = 0.8633
```

The p-value (0.8633) is greater than 0.05 which means we do not reject the null hypothesis hence our data is now homoscedastic.

### 3) Autocorrelation

**To test the presence of autocorrelation we used Durbin-Watson Test**

$H_0$ :  $\rho=0$ ; there is no autocorrelation

$H_1$ :  $\rho \neq 0$ ; there is autocorrelation

Level of significance= 5%= 0.05

```
> durbinwatsonTest(model4)
lag Autocorrelation D-W Statistic p-value
1      0.1870883      1.621206      0
Alternative hypothesis: rho != 0
```

The p-value is zero and d value is 1.62 ~ 2 which implies autocorrelation is not present.

### 4) Multicollinearity

**To test the presence of multicollinearity we used Variation Inflation Factors**

```
> vif_values2=vif(model4)
> vif_values2
data$`home ft2`  data$`lot ft2`
      1.079278      1.079278
```

Since both the vif values are less than 10 we can say that multicollinearity is not present.

**Recommendations:**

Given the non-normality, exploring alternative models such as Poisson regression might be considered. The Poisson regression model is expressed as:

## Poisson Regression Analysis

### The Poisson model

$$\Pr(Y = y) = \frac{e^{-\lambda V} (\lambda V)^y}{y!}$$

In this model  $E(Y) = \lambda V$ , and we typically assume

$$\begin{aligned}\ln[E(Y)] &= \ln(\lambda V) = \ln(\lambda) + \ln(V) \\ &= \beta_0 + \beta_1 X_1 + \dots + \beta_k X_k + \ln(V) .\end{aligned}$$

That is, we assume

$$\ln(\lambda) = \beta_0 + \beta_1 X_1 + \dots + \beta_k X_k .$$